

Looking into the future,
what can we learn about hacking
in science-fiction?



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Summary

Why imaginaries are so interesting?

Macro analysis

What does science fiction tell us about hacking?

Science fiction versus facts, any differences?

So, how do we use science fiction?

What are tomorrow's projections?

Everyone knows what imaginaries are.

**Few people realize how much it
shapes us.**

Why Imaginaries?

Methodological approach

What's the link between "science fiction" and "science facts"? A feedback loop

Imaginaires displays usages and technologies

Technologies, designers & engineers
are inspired by what is displayed and
work in order to produce the stuff in
real life



R/D displays forthcoming technologies

Authors speculate regarding the
consequences and usages of the R&D

Methodological approach

What's the link between "science fiction" and "science facts"?



YouTube: search for « Slaughterbots » by DUST

International Public Opinion (2015) 48, 105-120
Does Science Fiction Affect Political Fact? Yes and No: A Survey Experiment on "Killer Robots"
Kevin L. Smith and Kristen R. Soderstrom
University of Massachusetts Lowell

It's the End of the World and They Know It: How Dystopian Fiction Shapes Political Attitudes
Published online by Cambridge University Press: 26 November 2014

Rethinking the Political / -Science- / Fiction Nexus: Global Policy Making and the Campaign to Stop Killer Robots
Chad Carpenter

The messages simply generate awareness of their problems by playing on the cognitive dissonance between science "fiction" and political "fact". Popular culture is used strategically to "hook" advocacy targets into a conversation that then remains rooted in the substance of the real world. Cultural resources thus enable and inform activists' activities vis-à-vis their targets, be they stakeholders, allies or opponents.

We thus obtain proof that the message has an impact: the consumption of more frightening and apocalyptic killer robot stories correlates with greater opposition to autonomous weapons.

The results of three experiments are striking: we find consistent evidence that dystopian narratives reinforce the willingness to justify radical - particularly violent - forms of political action. Yet we find no evidence for the conventional wisdom that they reduce political trust and effectiveness, showing that the effects of fiction may not be what they seem, and underlining the need for political scientists to take fiction seriously.

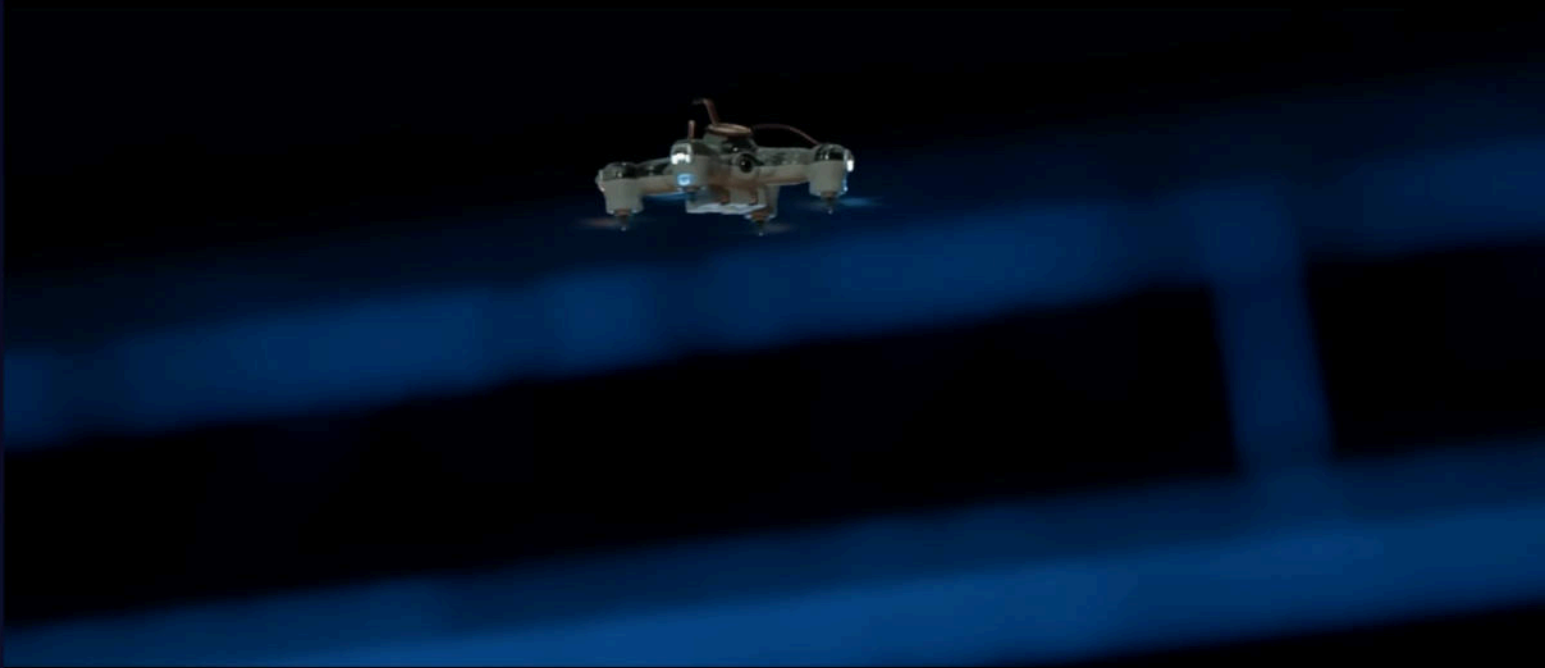


Customer pilots directed almost 3,000 precision strikes last year we're super proud of it.



It allows you to separate the bad guys from the good. It's a big deal.





The stochastic motion is an anti-sniper feature.

Wide Field Cameras

Face Recognition

Tactical Sensors



it has cameras and sensors. And just like your phones and social media apps, it does facial recognition.

Wide Field Cameras

Face Recognition

Tactical Sensors

Shaped Explosive



This is how it works?





Now that is an airstrike of surgical precision.



LATEST: ATTACK ON SENATE

DEADLY DEVICES BREACHED SECURITY

SDN

which officials are describing as some kind of automated attack, which killed 11 US senators at the Capitol building?





LIVE

LATEST: 8,300 STUDENTS KILLED. 12 UNIVERSITIES HIT

The search for a motive is apparently turning to social media and a video shared by the victims



I'm Stuart Russel, a professor of computer science at Berkley. I've worked in AI for more than 35 years.

It shows the results of integrating and miniturizing technologies that we already have.

It's potential to benefit humanity is enormous, even in defense.

But allowing machines to choose to kill humans will be devastating to our security and freedom.

Thousands of my fellow researchers agree.

We have an opportunity to prevent the future you just saw, but the window to act is closing fast.

Imaginations of the future
shape research and politics
agendas in the present

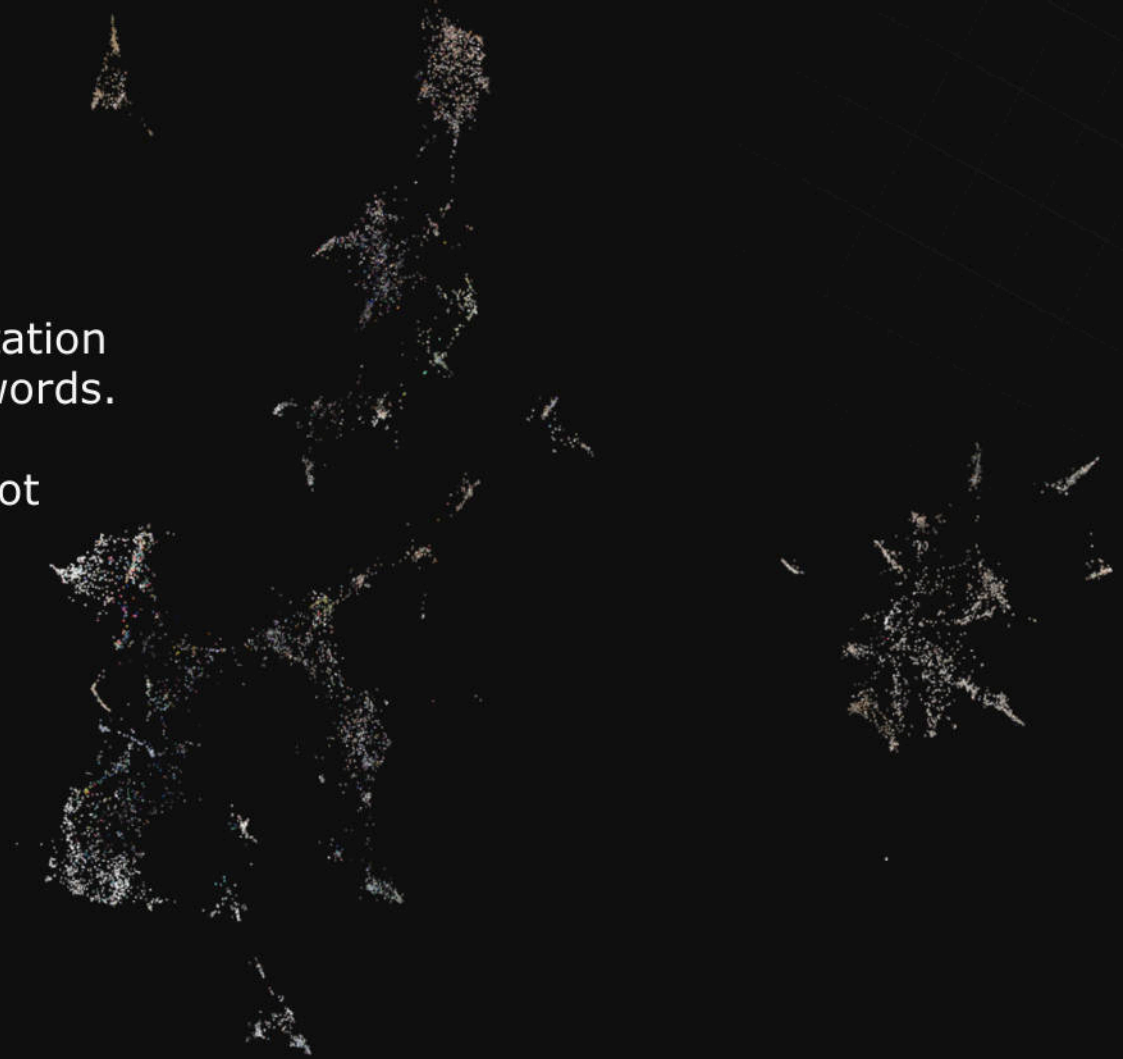


Macro analysis of **Hacking** in present imaginaries

Representation of « Hacking »

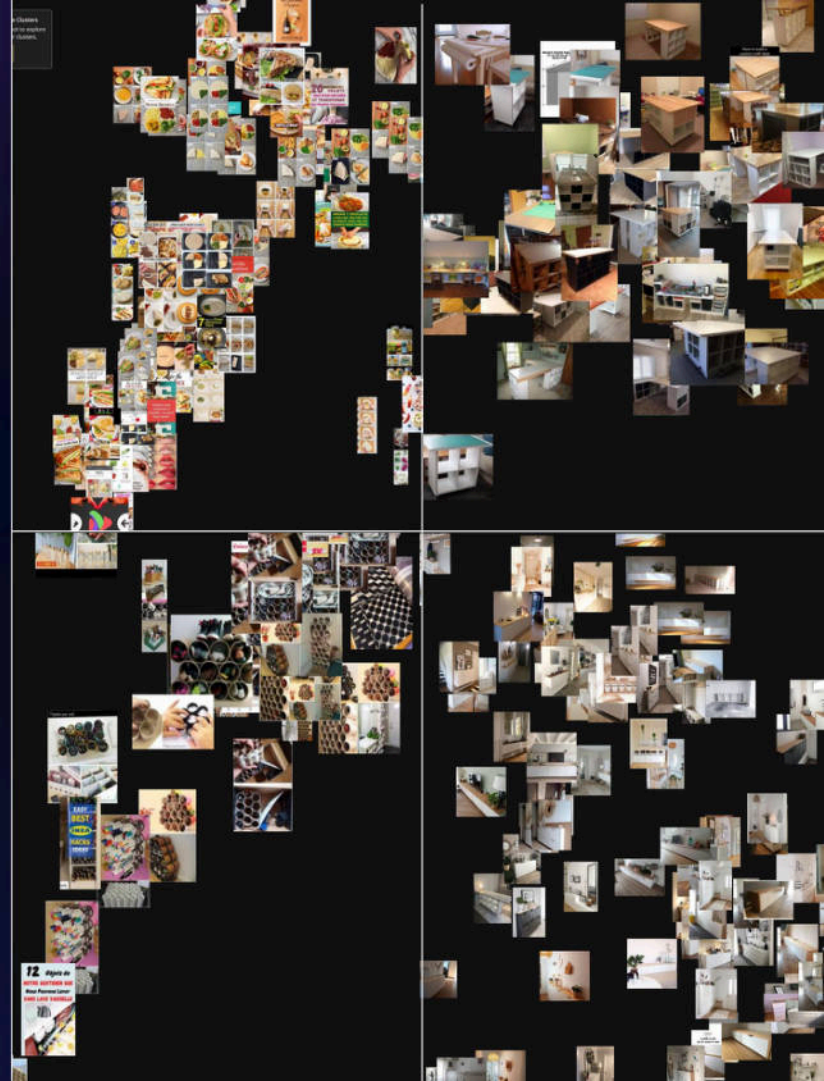
We crawled Pinterest, ArtStation
and Shutterstock with keywords.

We used Yale DHLab's PixPlot
to map images.

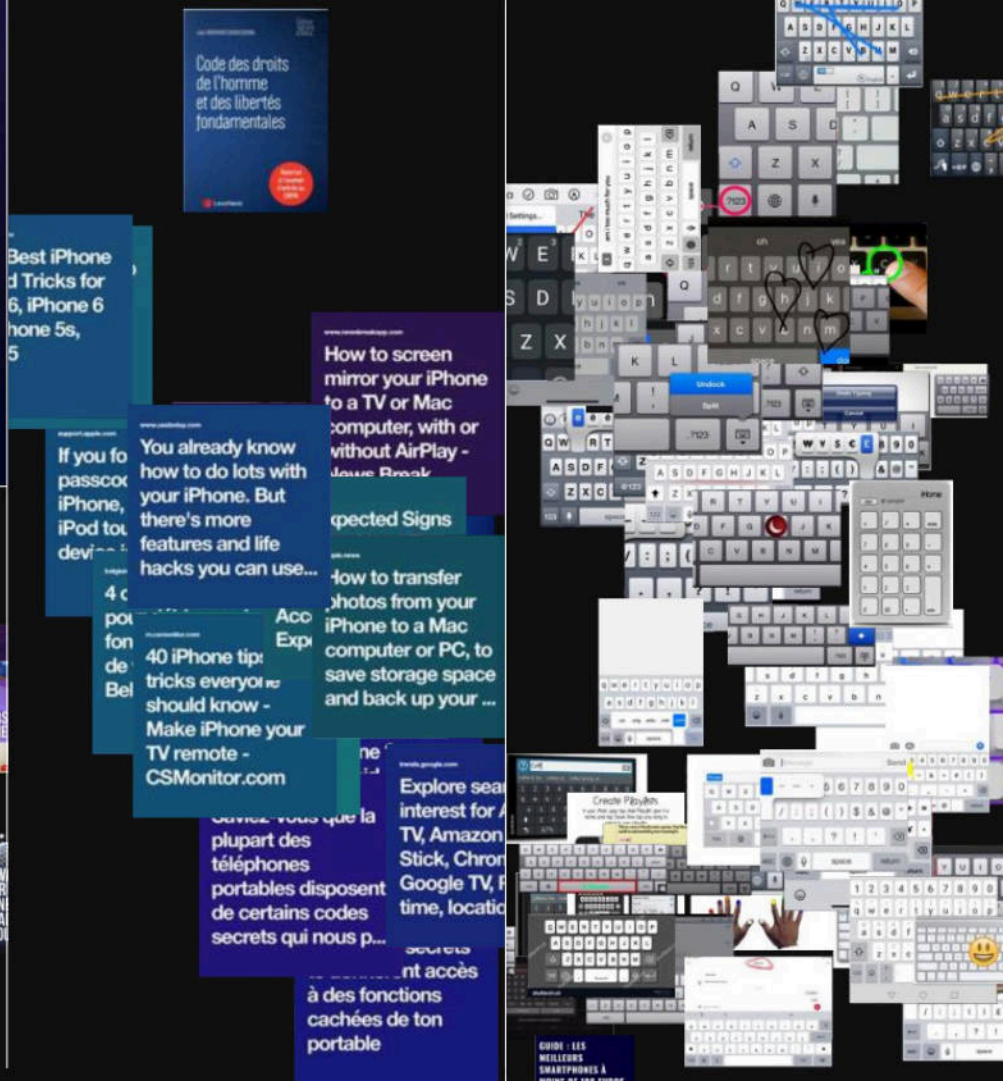
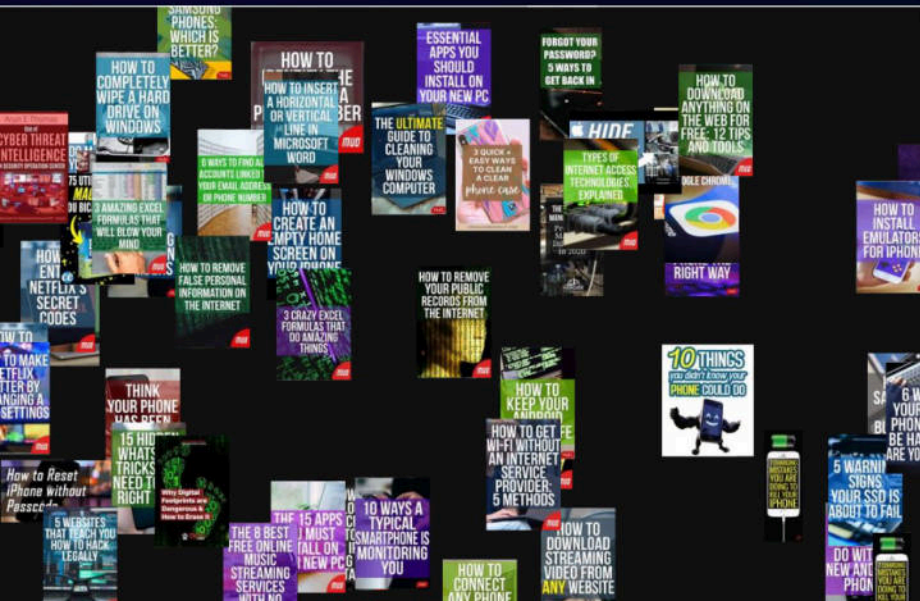


Representations of « hacking »

Appropriation of the term



Simplistic visions of hacking

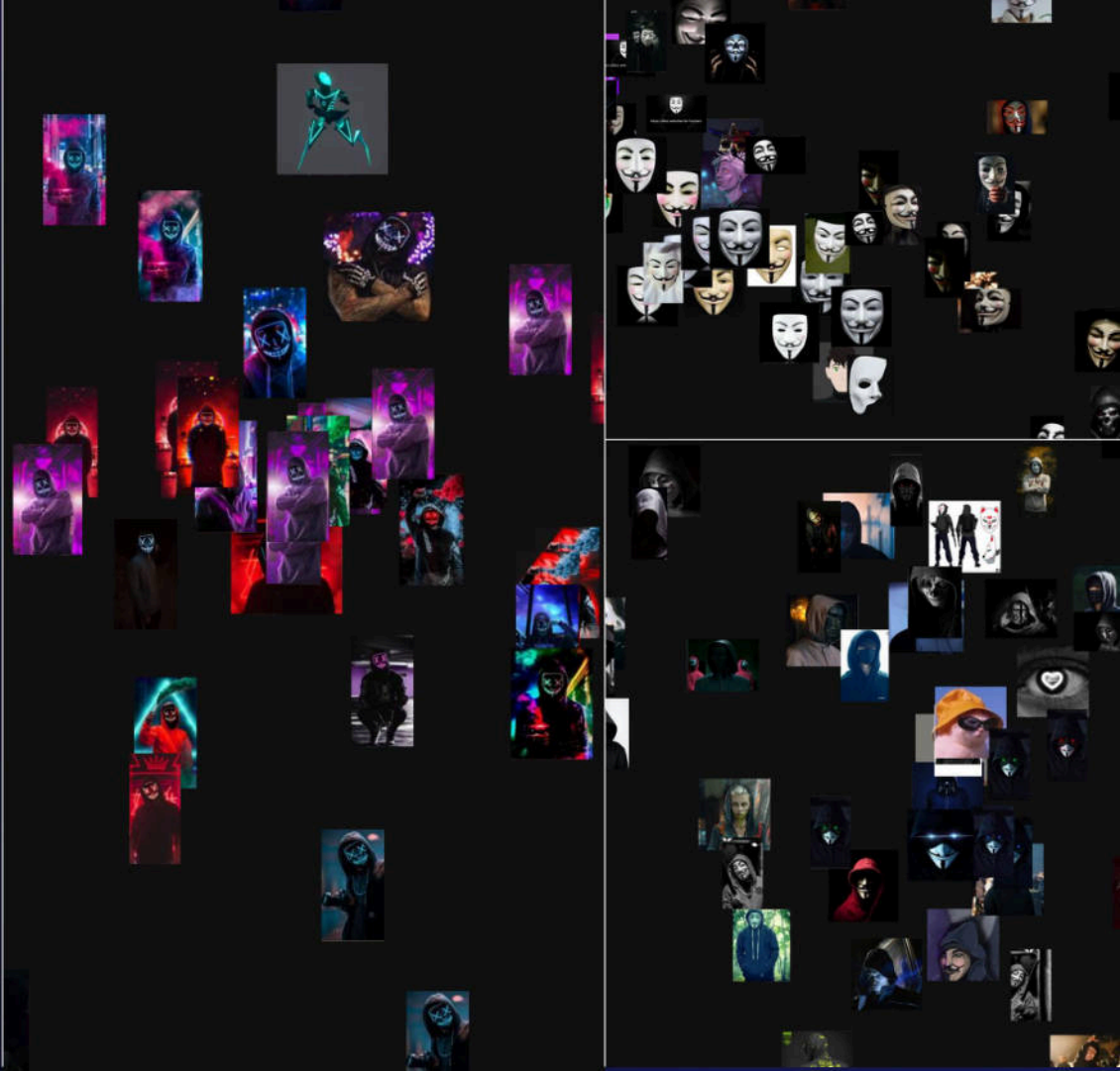




Representation of « Hacker »

Representations of « hacker »

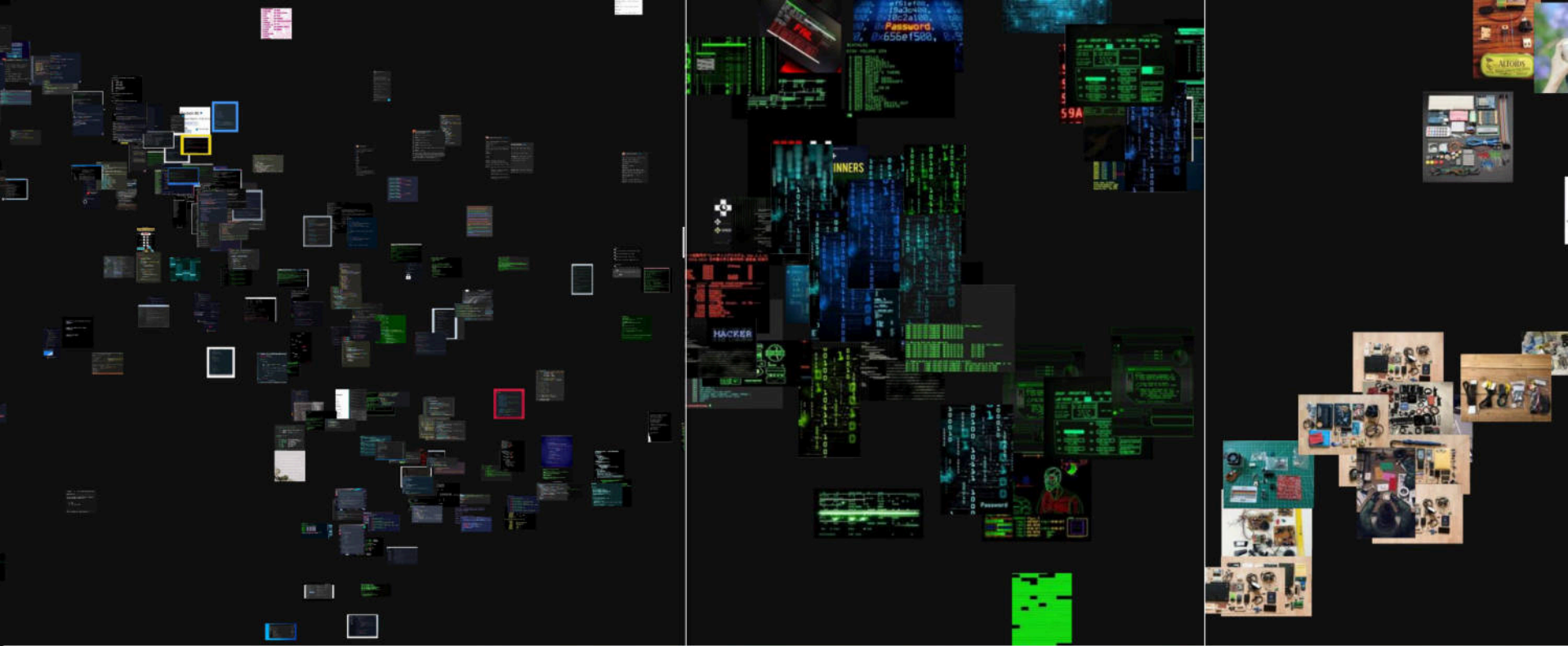
Strong archetypes



Got it



Strong archetypes



Representations of « hacker »

A more technical dimension

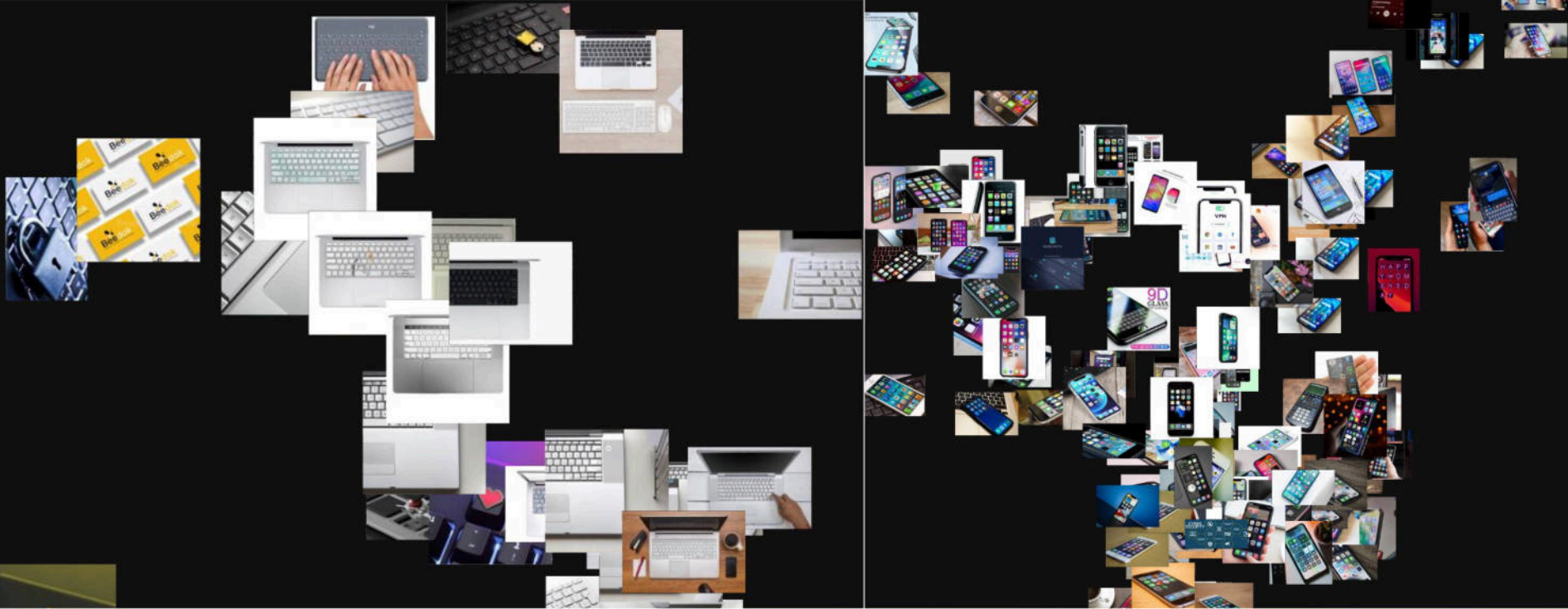
The background of the slide is black with a faint, light gray grid pattern. Two large, abstract shapes composed of numerous small, multi-colored dots (white, blue, yellow, and red) are positioned on the left and right sides of the slide. The shape on the left is elongated and somewhat irregular, while the shape on the right is more compact and rounded. In the center of the slide, the text "Representation of « Cybersecurity »" is displayed in a white, sans-serif font. The word "Representation of" is on the top line, and "« Cybersecurity »" is on the bottom line, with the word "Cybersecurity" in a blue color.

Representation of
« Cybersecurity »



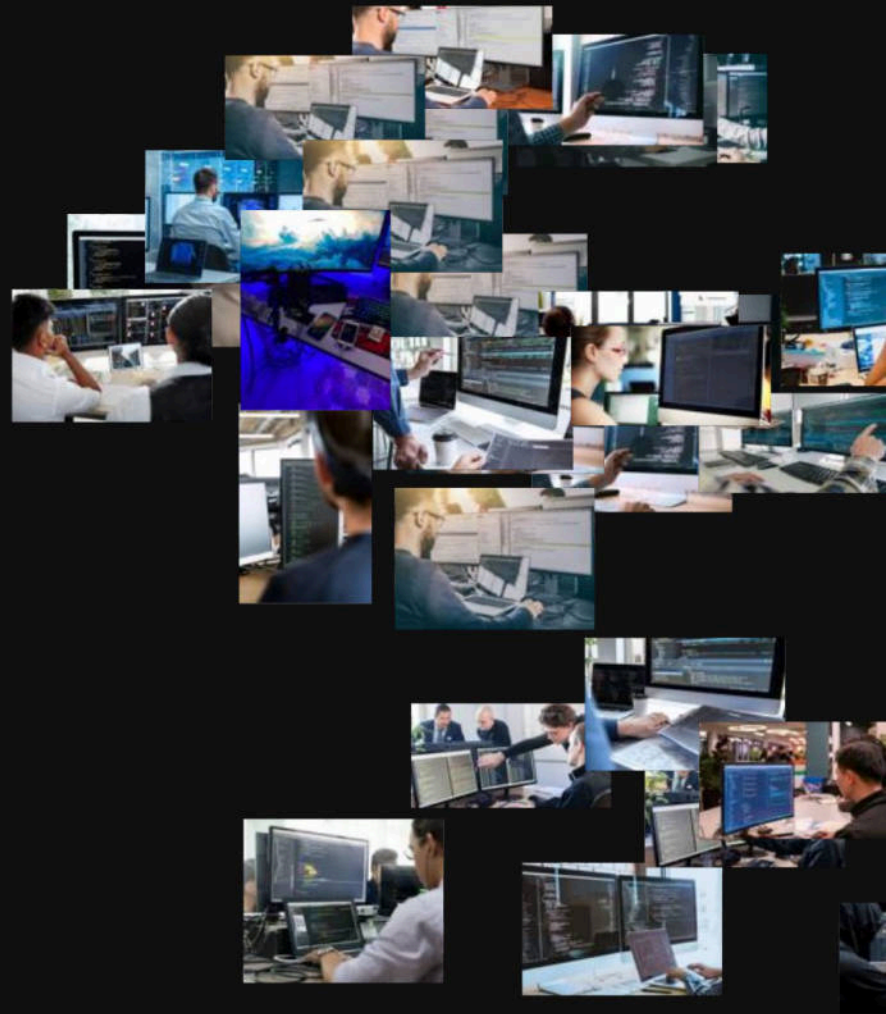
Representations of « cybersecurity »

The eternal padlock



Representations of « cybersecurity »

Devices



Representations of « cybersecurity »

Humans

Caricatural representations



Using representations for a purpose:

The IDEO and HP contest

01

Humanizing representations

- Go back to basics and explain that it's humans who compromise infrastructure, not circuit boards.
- Give a face to the "hacker hoodie".

02

Being accessible

- A highly complex subject that's not easy to understand.
- Engage citizens by presenting simple, accessible elements.

03

Sounding the alarm without being alarmist

- Narratives based on fear and catastrophism do not allow us to focus on the causes of the problems, nor to approach the problem in a systemic way.

04

Accuracy is credibility

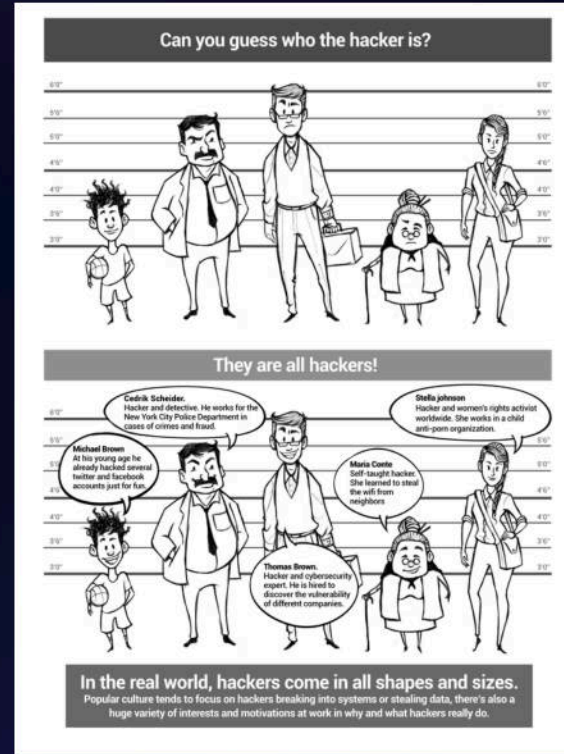
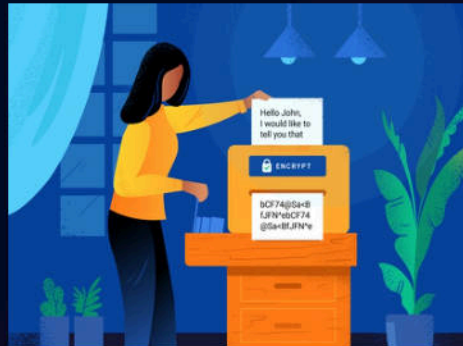
- Visuals must explain how things really work
- The difficulty lies in having imagery that is accessible to all, but that shares real technological fundamentals.

05

Metaphors for complex issues

- How to make the invisible visible ?
- The main challenge here is to make issues such as encryption and privacy tangible.

Some of the results, trying to go further than « flatpacks futures » of Hollywood



Quantitative analysis

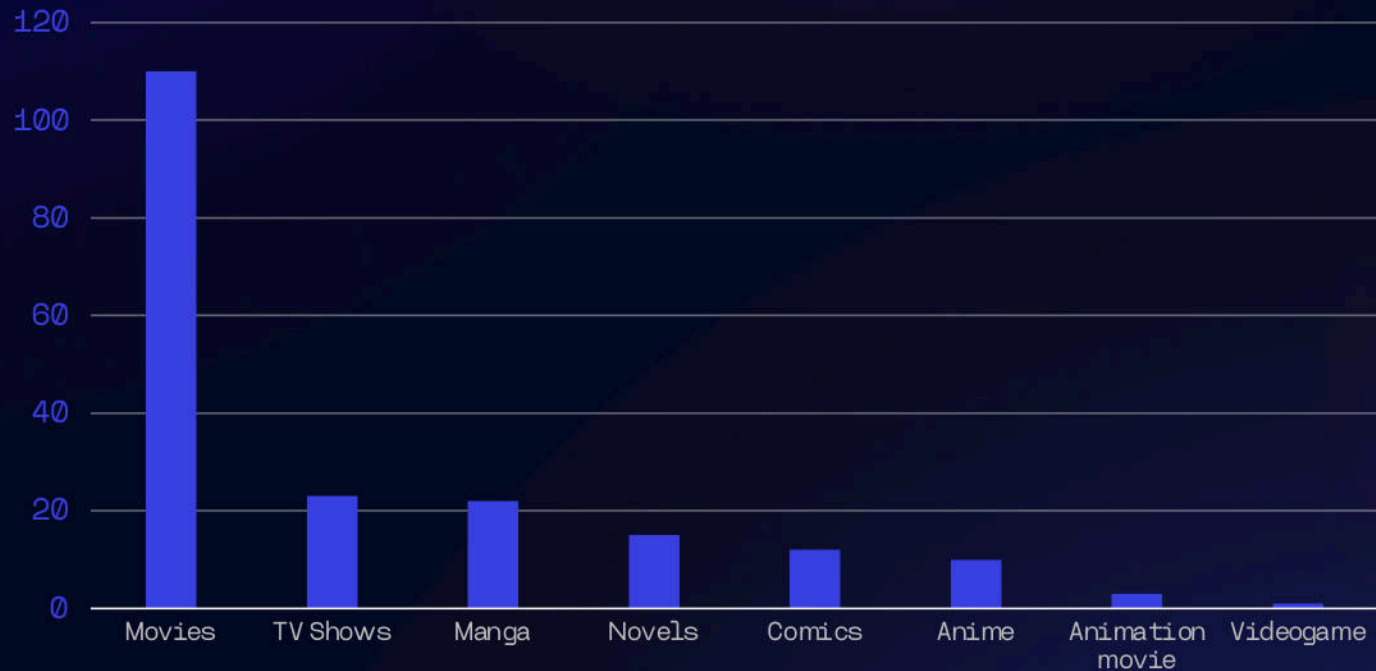
**What does science
fiction tell us ?**

191 entries for fictions with 5 major variables

Title	Year	Media	Attack surface	Target	Hacker_type
The Italian Job	2003	Movie	Control center	Traffic Light	Independant
Neuromancer	1984	Novel	NaN	Data, AI	Independant
CSI: Cyber	2015	Series	Car	Car, Human Life	Independant
Homeland Season 2	2012	Series	Pacemaker	Pacemaker	Mafia, Terrorist Group
Wargames	1983	Movie	Control center	Missile Systems	Independant
Watch Dogs 2	2016	Videogame	Control center; Infrastructure	City Infrastructure, Traffic Light, Data, Money, Person	Independant
Johny Mnemonic	1995	Movie	Brain	Brain	Independant, Company, Mafia, Terrorist Group, AI or Robot
Ghost in the Shell	1989	Manga	Control center; Brain; Armor	Brain, Exoskeleton, Control Center	Independant, Hacker Group, State, AI or Robot, Company

The database

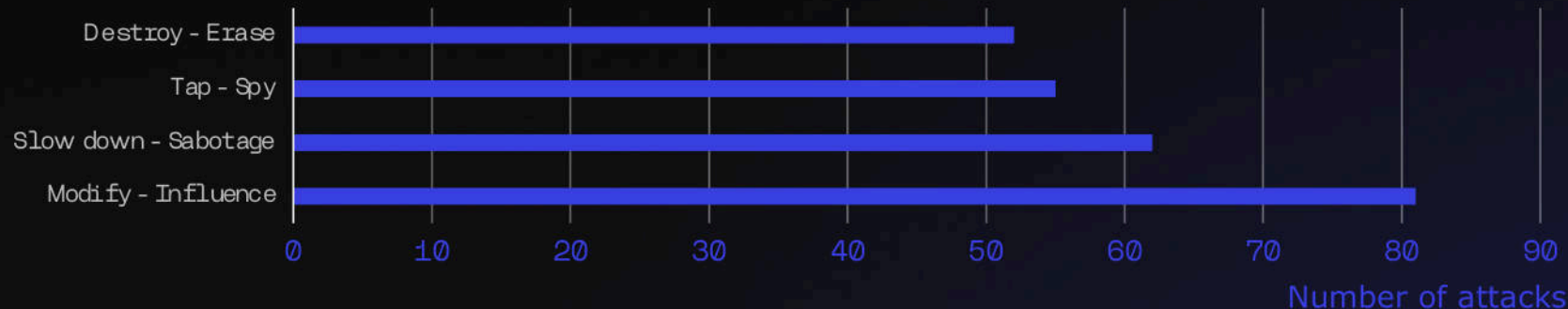
Distribution of media types



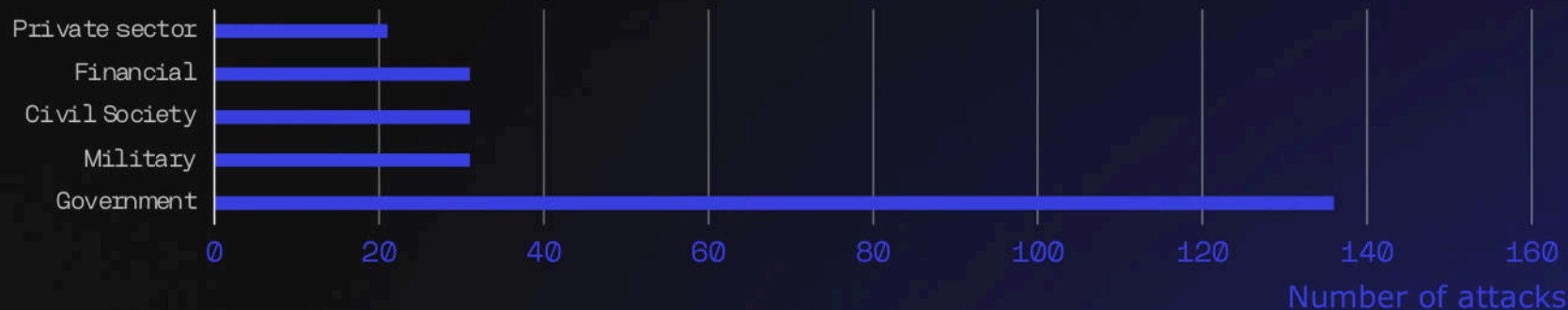
The database

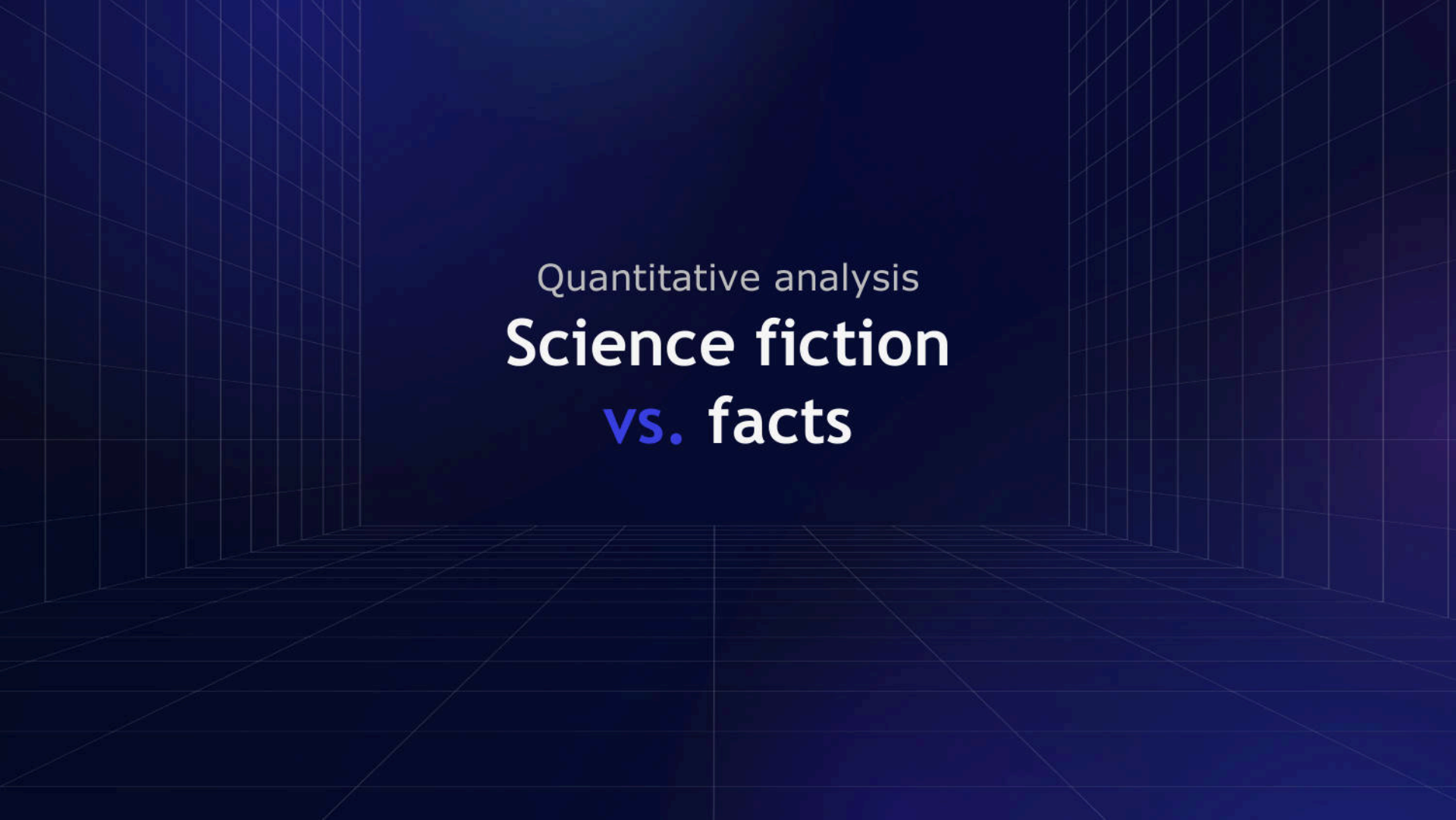
What distributions for [quadrants] and [targets] ?

Distribution of Quadrants for fiction



Distribution of Targets for fiction





Quantitative analysis

Science fiction **vs.** facts

976 entries for Reality

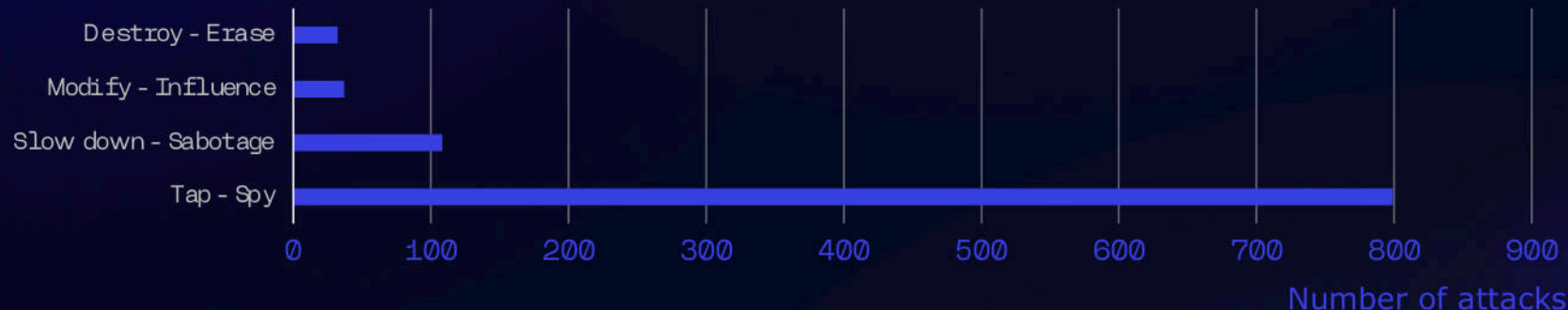
3 sources :

- US Council of Foreign Relations database
- The Cybercrime History Data Line by Robert Herjavec
- Interviews

Attacks	Date	Targets	Quadrant
French Telegraph System	1834	Financial	Tap - Spy
Switchboard Hack	1870	Private Sector	Modify - Influence
Early Telephone Calls	1878	Private Sector	Modify - Influence
Wireless Telegraphy	1903	Private Sector	Modify - Influence
Military Codebreaking	1939	Military	Tap - Spy

Reality : Distribution for quadrants and targets

Distribution of Quadrants for Reality



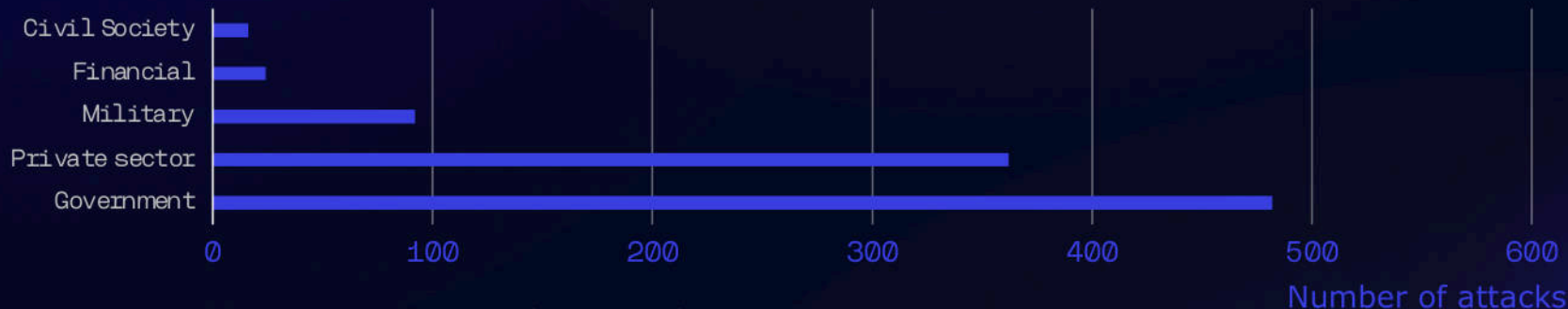
Distribution of Targets for Reality



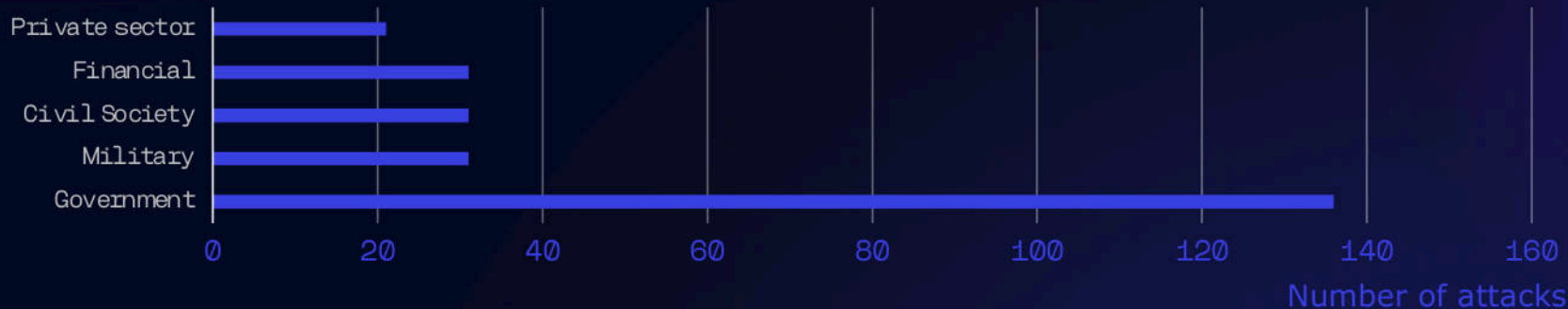
Targets: Reality compared to Fiction

Mind the bias : Reality dataset is declarative

Distribution of Targets for Reality



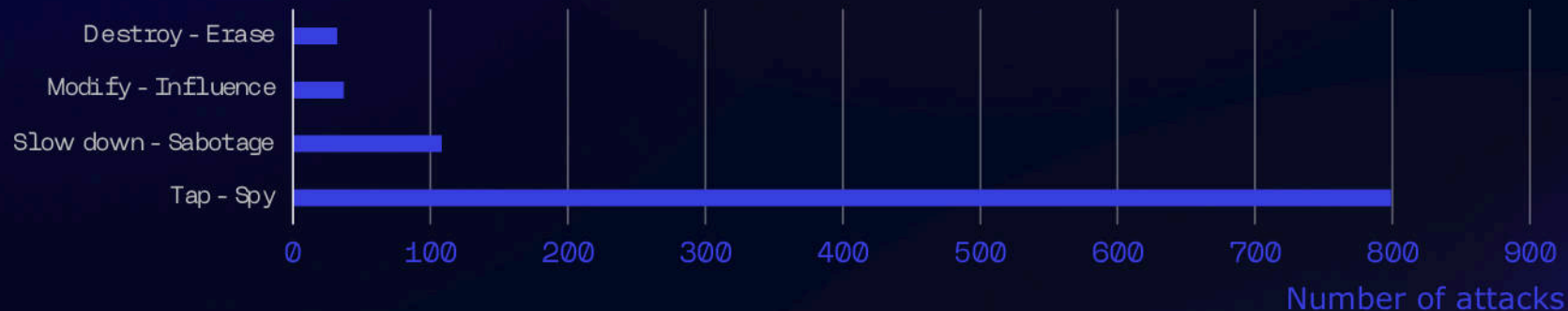
Distribution of Targets for Fiction



The database

Comparison [reality] * [fictions] for [quadrants]

Distribution of Quadrants for Reality



Distribution of Quadrants for fiction



**But there is an inflexion point in
reality around 2008/2010**



2008
Explosion of
Baku-Tbilisi-
Ceyhan oil
pipeline

Quick stop on 2 examples of physical world threats/attacks

- Hacking a Nuclear Power Plant
- Hacking a Pacemaker

Hacking a nuclear power plant

Fiction

1984 Hide and Seek movie
A college kid break into a
canadian nuclear facility



ATOMIC POWER CANADA
0900 CONGERENCE ROOM A
INVESTIGATION COMMITTEE
SECURITY CODE 10

Reality

2010 Stuxnet attack destroy
centrifuges of Natanz nuclear
plant



Hacking a Pacemaker

Fiction

Dec 2012 Homeland TV Show
Murder of the US Vice President



Reality

2008 D. Halperin's paper « Pacemaker and Implantable Cardiac Defibrillators: Software Radio Attacks and Zero-Power Defenses »

2013 VP Dick Cheney remove the wireless features of his pacemaker

« I found it credible »

« It's an accurate representation of what's possible. »

2017 Recall of half a million

« hackable » pacemakers



2018 Pacemaker vulnerability demonstrated at BlackHat (Jonathan Butts & Billy Kim)
Threat reported to manufacturer in 2016

Ready for the last part?

Let's finish on a positive vibe!

How can we use
science fiction?

*example of a qualitative
innovation track*

Cartographic representations



Kaspersky



Akamai



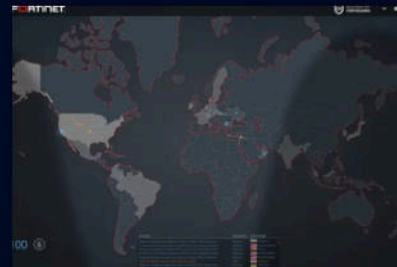
Arbor



Checkpoint



Fireeye



Fortinet



Norse



Trend Micro

Cartographic representations

Strong science fiction references

Kaspersky



Fireeye



Norse



Colossus: The Forbin project
(1970)

Cartographic representations

Strong science fiction references

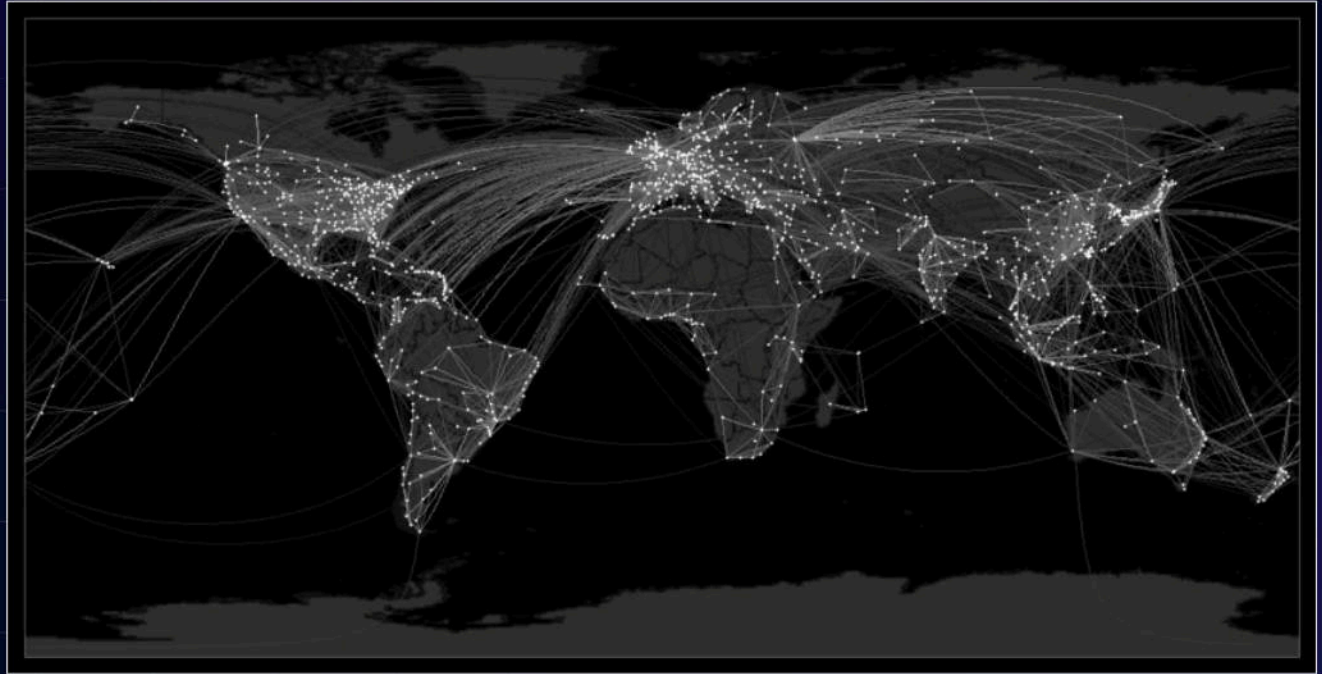
Kaspersky



Fireeye



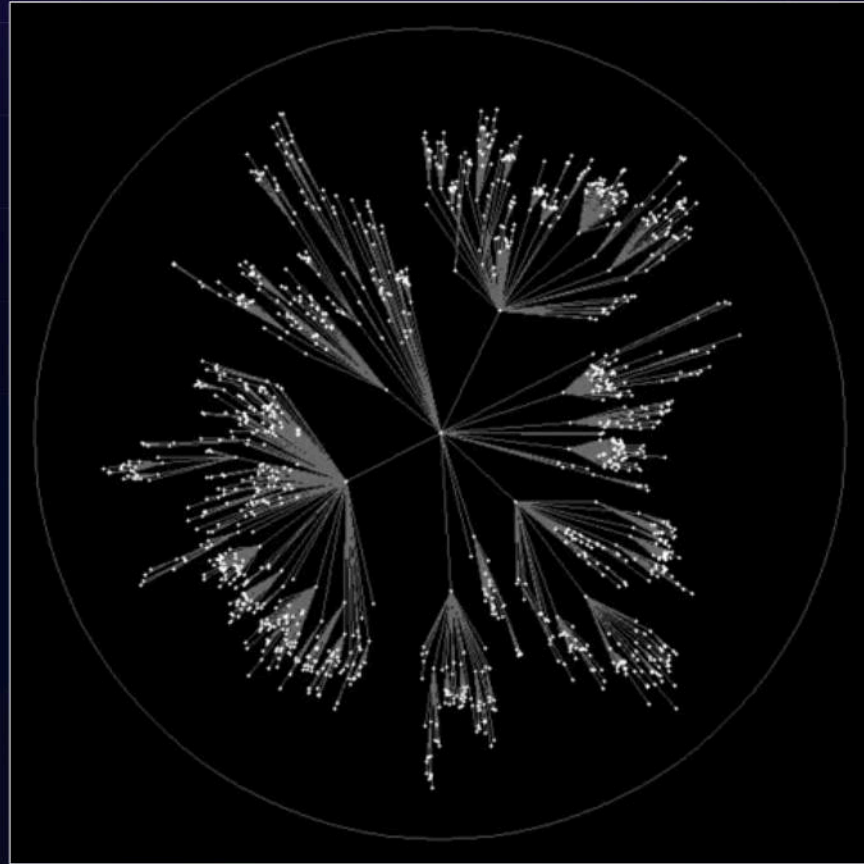
Arbor



**Professor Dirk Brockmann's modeling system study for pandemic
2013 : Wan Epi Paphos**

Cartographic representations

Strong science fiction
references



**Professor Dirk Brockmann's modeling system study for pandemic
2013 : Wan Epi Paphos**

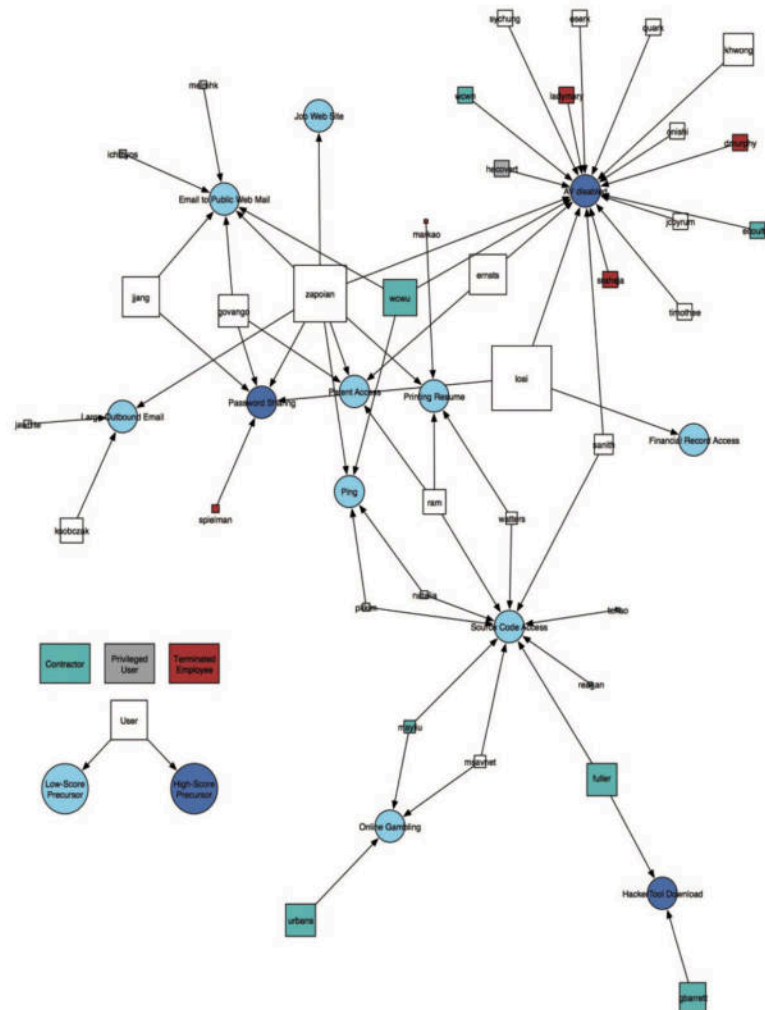
Cartographic representations

Critical visualization: a case for rethinking how we visualize risk and security

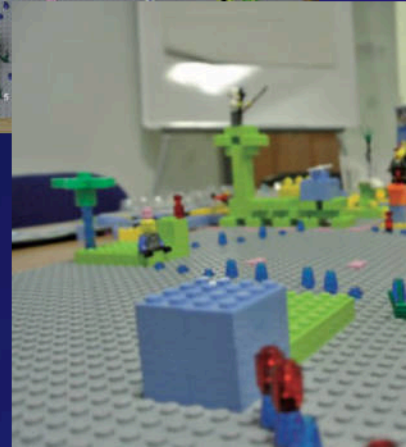
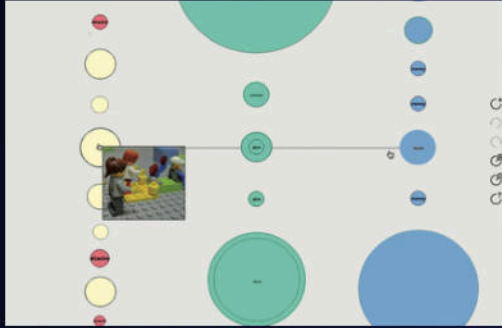
Peter Hall^{1,*} Claude Heath² and Lizzie Coles-Kemp²

"The predominant mode of visualization in the security field comes from a statistical and probabilistic approach that perpetuates a particular way of looking at the problem and is based on a relatively thin narrative of cybersecurity."

For example, Raffael Marty's "Applied Security Visualisation" uses "progressive disclosure" for the iterative elimination of "outliers", based on the analysis of network nodes that generate traffic with large packets and determining whether they reveal suspicious distribution patterns. In this way, Marty arrives at a suspect botnet controller. The visualization, according to Marty, is worth a thousand log records.



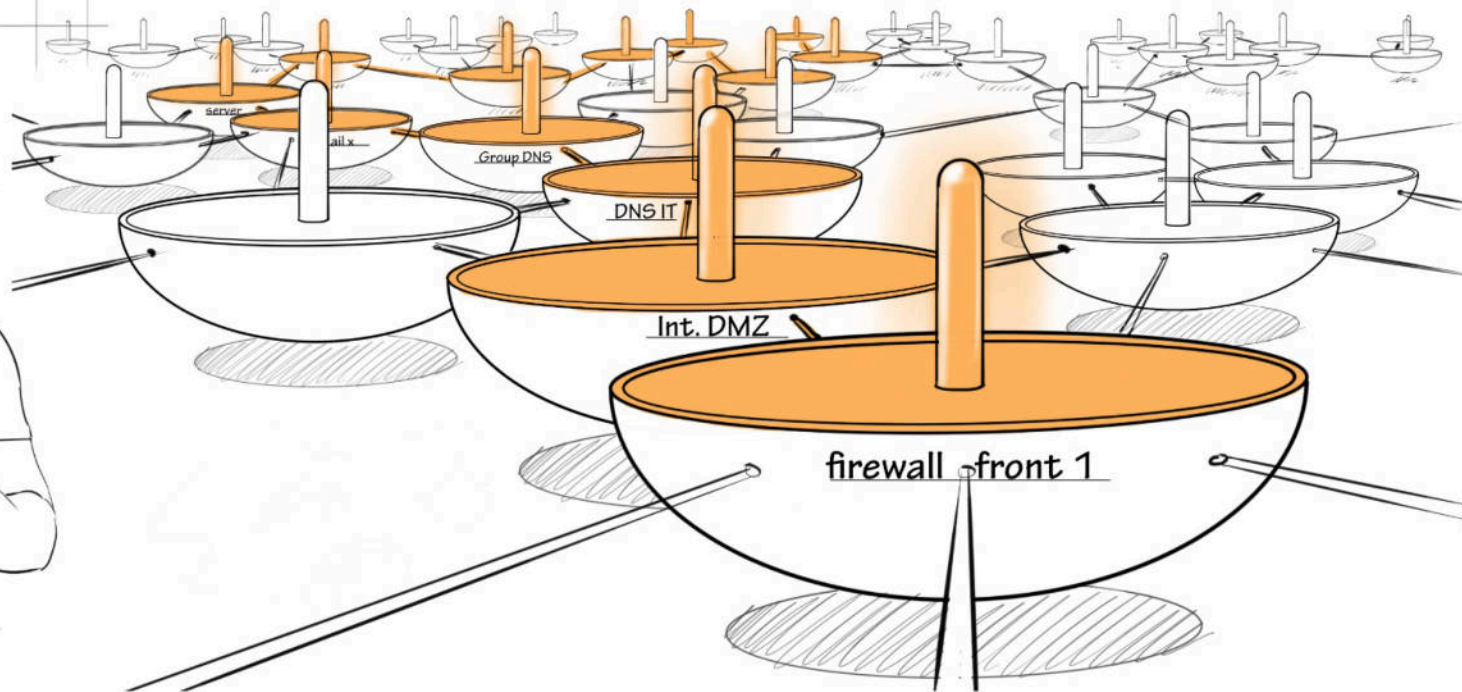
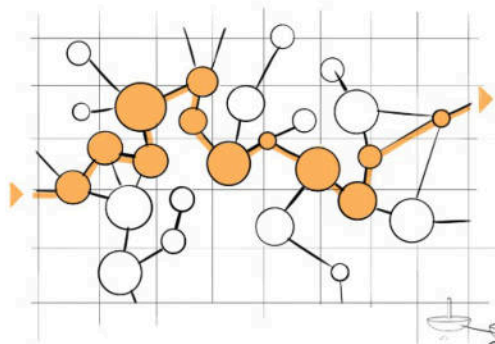
Cartographic representations

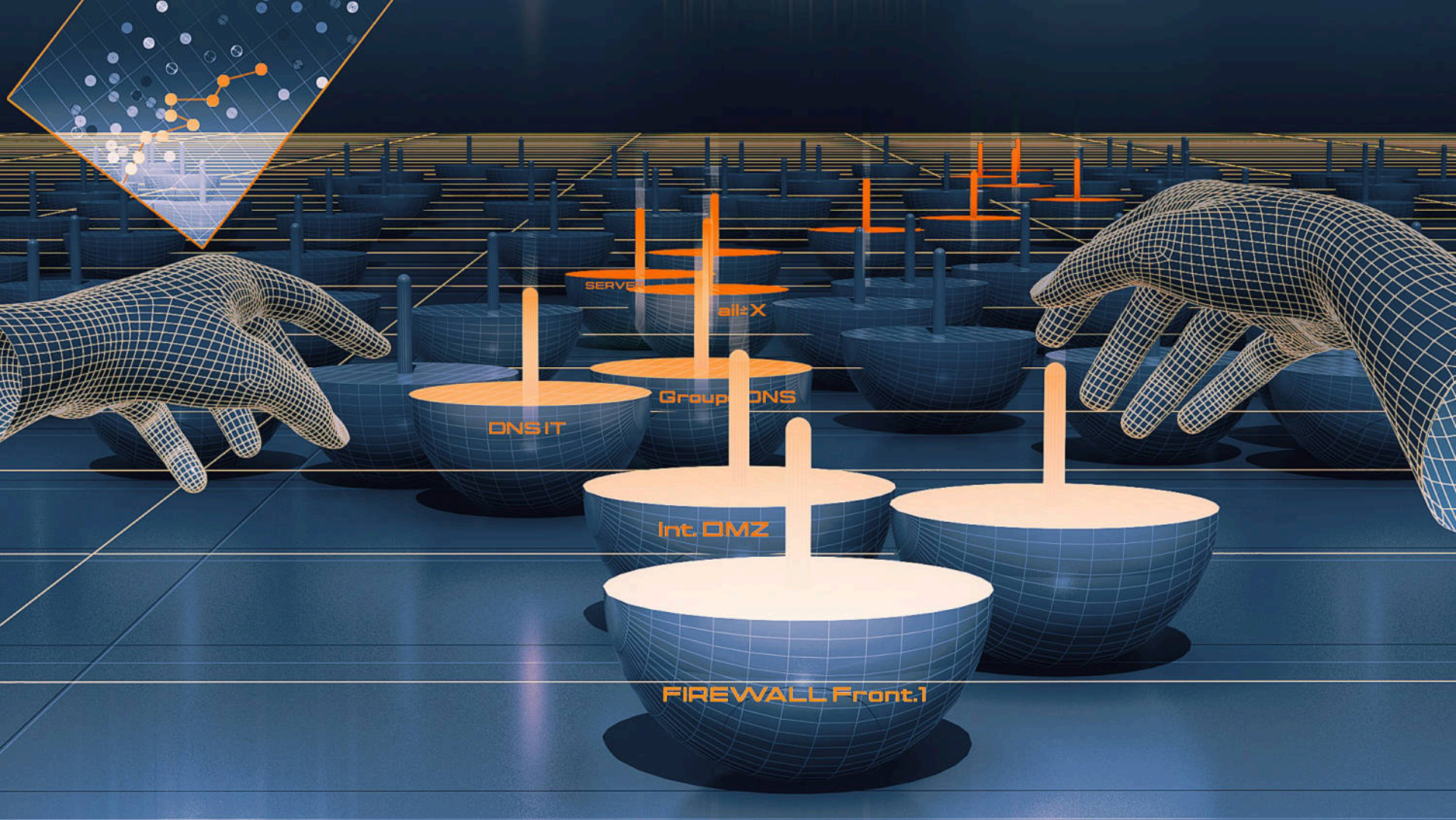


Cartographic representations

Strong science fiction references







FIREWALL Front.1

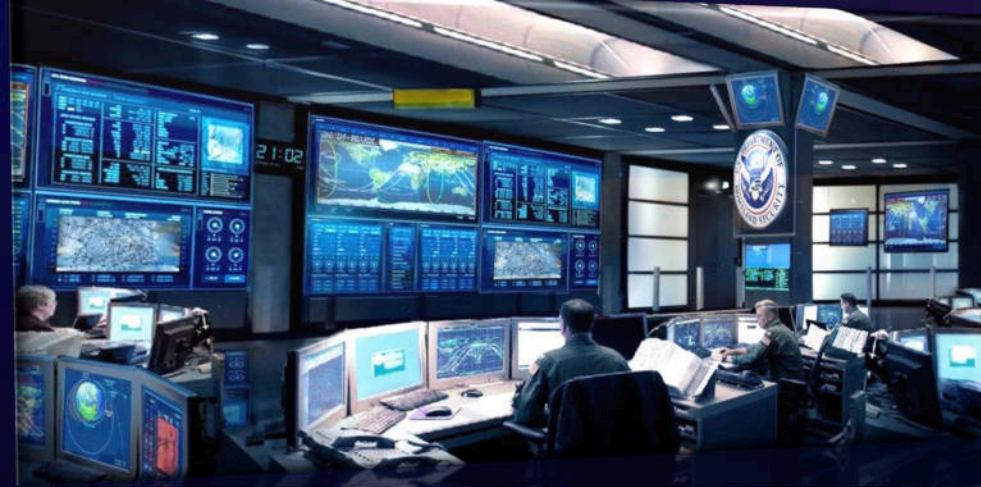
Int. DMZ

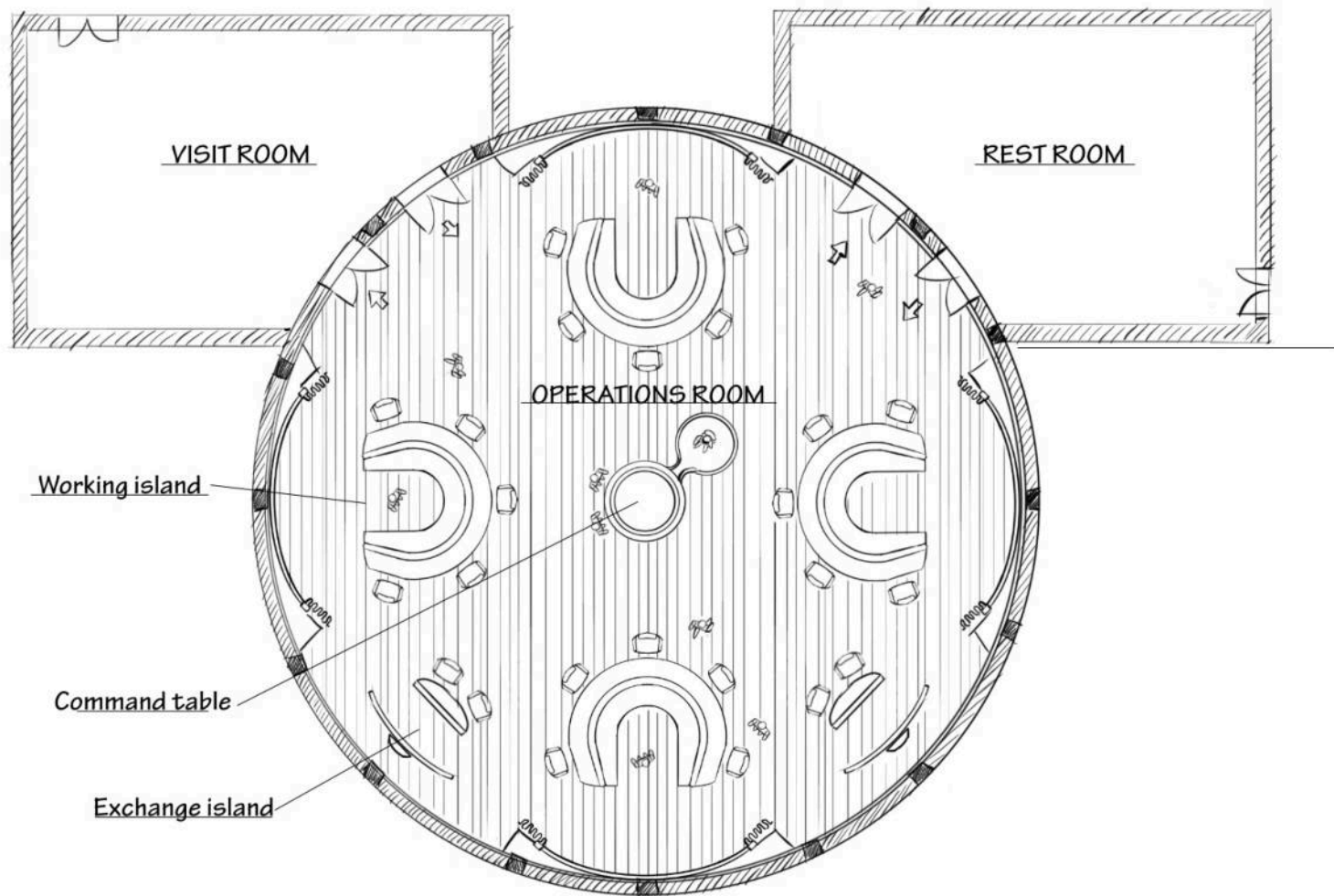
Group DNS

DNS IT

SERVER

all X





Panoramic
interactive board

Confidential part

Acoustic and lighting
environment with green plants

audio - video capture

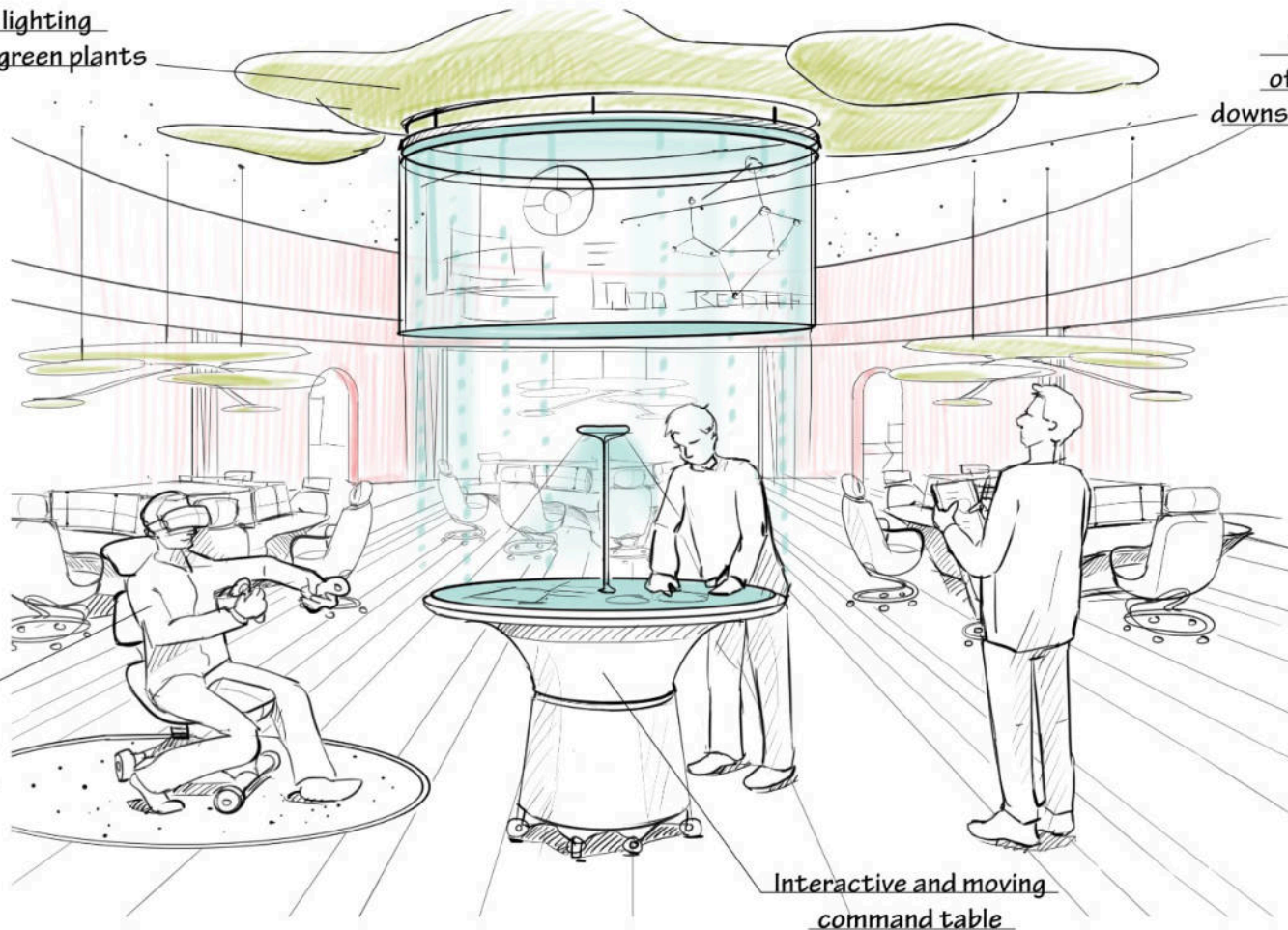
horseshoe desk

Working island



Acoustic and lighting
environment with green plants

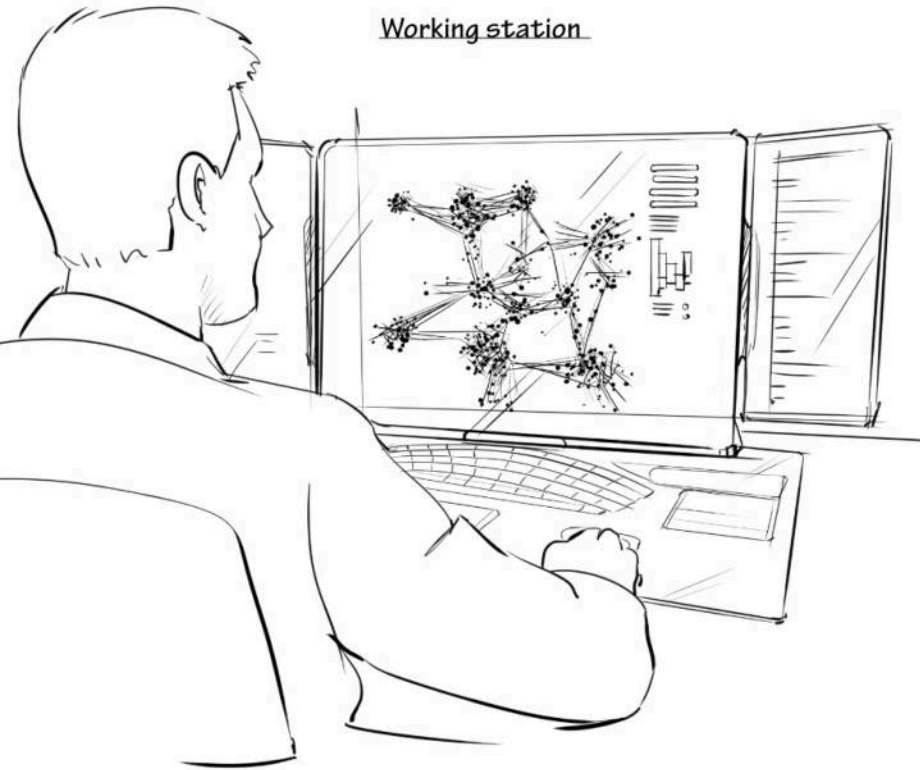
Viewing dome
of upstream and
downstream informations



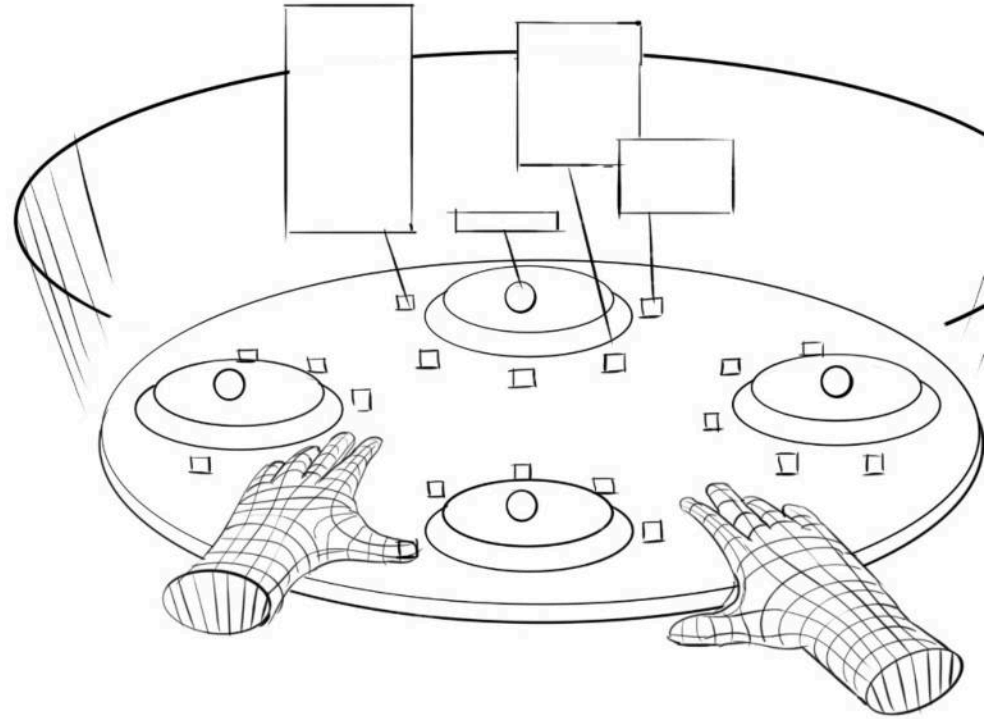
VR Pilot station
of operations

Interactive and moving
command table

Working station



VR Pilot station
of operations







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